Karnaugh Maps

Contemporary Digital Electronic Systems — Chapter 6

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Why This Chapter Matters

Boolean algebra simplifies by rewriting rules. It is error-prone and does not show whether the result is the smallest. A Karnaugh map is a graphical procedure that finds a minimum SOP (or POS) directly from the truth table, for functions of up to four variables.

After this chapter, you can:

1. Lay out a 2-, 3- and 4-variable K-map in Gray-code order.
2. Transfer a truth table into a K-map and back.
3. Identify prime and essential prime implicants.
4. Extract a minimum-cost SOP (or POS) expression.
5. Use don’t-cares to reduce the expression further.

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Why is the K-map laid out in Gray-code order, not normal binary order?

What property of adjacent cells does that layout preserve? Keep this in mind as we build the maps.

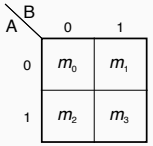
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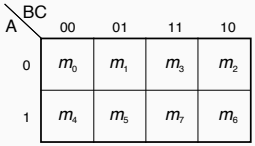
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What Is a Karnaugh Map?

- A K-map (or Veitch diagram) is a truth table drawn as a 2-D grid. Each cell is one truth-table row.
- An n-variable map has 2^n cells.
- Best for small problems (up to four variables); impractical beyond that.
- Simpler expressions mean a simpler circuit, with more reliability and speed and lower cost.
- Two adjacent 1s always group: they differ in one variable x, and $x \cdot t + \overline{x} \cdot t = t$ by the unifying theorem, so x drops out.



2-variable map (m_0 to m_3)



3-variable map (m_0 to m_7)

Gray-Code Order Keeps Neighbours Adjacent

- Rows and columns follow Gray-code order, so the minterm indices of adjacent cells differ in exactly one bit.
- Adjacent cells differ in one variable. That variable can be eliminated when they are grouped.
- This answers the opening question.
- A 2-variable function has only $2^{2^2} = 16$ possible maps, so those maps have limited use.

A \ BC	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

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Filling in a K-Map

$F = AB + AC$

Convert to canonical SOP first:
$= AB(C + \overline{C}) + AC(B + \overline{B})$
$= AB\overline{C} + ABC + ABC$
$= \sum m(5, 6, 7)$

A \ BC	00	01	11	10
0	0 (0)	0 (1)	0 (3)	0 (2)
1	0 (4)	1 (5)	1 (7)	1 (6)

Prime numbers 0 to 7

Primes: 2, 3, 5, 7. So $F = 1$ at m_2, m_3, m_5, m_7.

A \ BC	00	01	11	10
0	0 (0)	0 (1)	1 (3)	1 (2)
1	0 (4)	1 (5)	1 (7)	0 (6)

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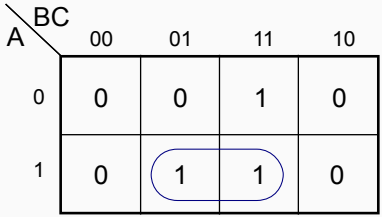
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Grouping Adjacent Cells

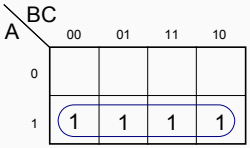
- Two adjacent minterms differ in one variable, so by the unifying theorem $A + \overline{A} = 1$ that variable disappears:

$$A\overline{B}C + ABC = AC(\overline{B} + B) = AC$$

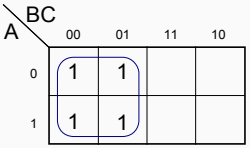
- A loop of 2^n cells eliminates n variables. Larger loops give simpler terms.
- Cells on opposite edges are also adjacent: the map wraps around.



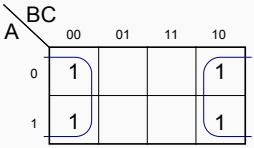
Example Loops (3 Variables)



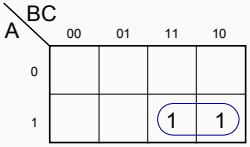
F = A



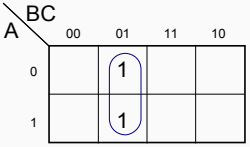
F = $\overline{B}$



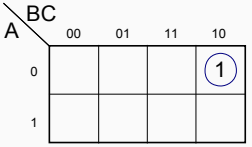
F = $\overline{C}$



F = AB



F = $\overline{B}\overline{C}$



F = $\overline{A}B\overline{C}$

- Only cells containing 1 are looped.
- The grouping for $F = \overline{C}$ shows the wraparound property.

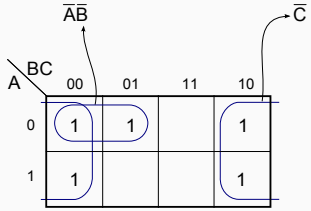
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K-Map Simplification Procedure

1. Place all minterms on the map and mark the cells containing 1.
2. Form rectangular loops of 2^n cells (1, 2, 4, 8, ...).
3. Always choose the largest possible loops.
4. A cell may belong to more than one loop if that gives a simpler result.
5. Continue until every 1 is in at least one loop.
6. Write one product term per loop and OR the terms together.

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Example: $F = \sum m(0,1,2,4,6)$



- Loop of four: m_0, m_2, m_4, m_6. A and B change, $C = 0$: term $\overline{C}$.
- Pair: m_0, m_1. Only C changes: term $\overline{A}\,\overline{B}$.

Result

$F = \overline{A}\,\overline{B} + \overline{C}$

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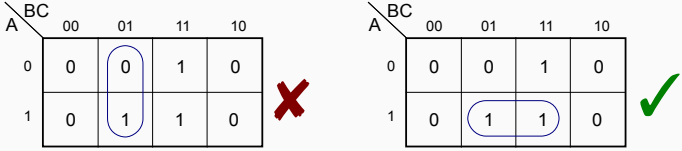
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6 Don’t Cares and POS

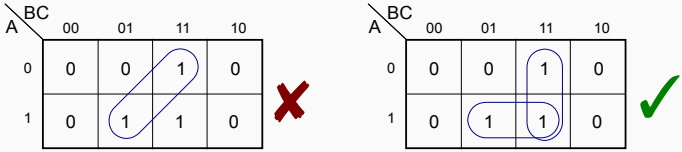
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Rules 1 and 2

Rule 1: No zeros allowed. A group may not contain a cell with 0: it would assert an output the truth table says is 0.

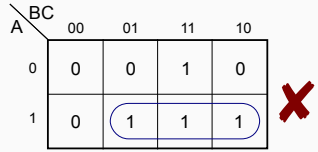


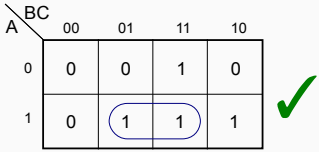
Rule 2: No diagonals. Adjacent cells differ in one variable, so it cancels; diagonal cells differ in two, so nothing cancels.



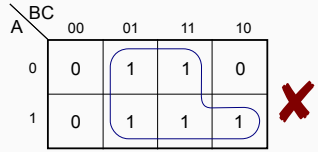
Rule 3: Only Powers of 2

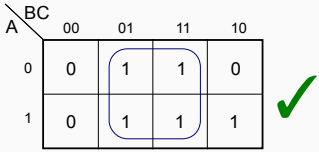
Rule 3. A group of 2^k cells drops exactly k variables, so sizes must be 1, 2, 4, 8, ...





A group of 3 or 6 cells is not allowed.

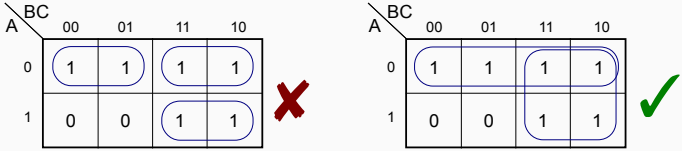




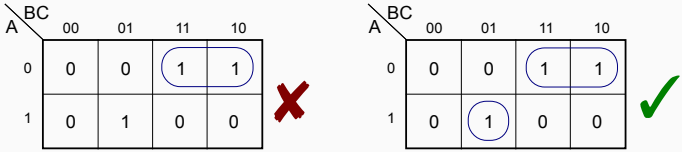
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Rules 4 and 5

Rule 4: Each group as large as possible. A smaller group is still correct but *not minimal*.

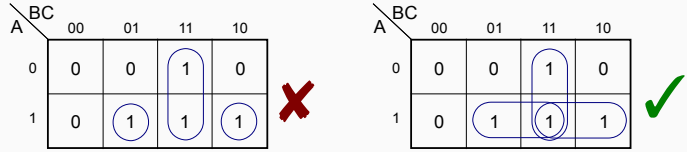


Rule 5: Every 1 must be in at least one group. An uncovered 1 would give output 0 for that input.

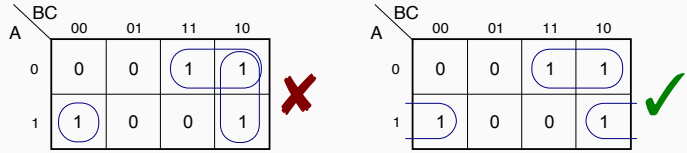


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Rule 6: Groups may overlap. Covering a cell twice is harmless ($A + A = A$) and often lets groups grow larger.

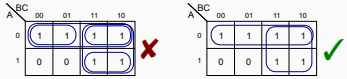


Rule 7: Groups may wrap around. Left edge with right edge; top with bottom.



Rule 8 and Prime Implicants

Rule 8: Fewest groups. A small group inside a bigger one adds an unnecessary gate.



- A prime implicant is a group of 1s obeying Rules 1–3 and 7 that cannot be enlarged (Rule 4).
- An essential prime implicant covers a 1 that no other prime implicant covers. It must appear in every minimum SOP.

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1. Circle all prime implicants.
2. Select every essential prime implicant. Leaving one out would leave a 1 uncovered (Rule 5).
3. If any 1s remain uncovered, add the fewest other prime implicants that cover them (Rule 8).

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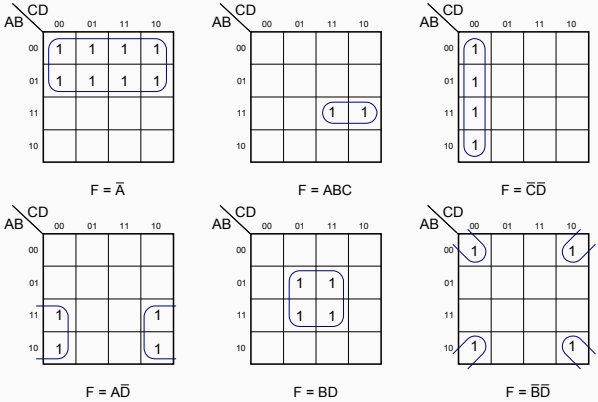
The Four-Variable Map

AB \ CD	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

- $2^4 = 16$ cells, numbered m_0 to m_{15}.
- Rows and columns are in Gray-code order, so the map wraps horizontally and vertically.
- The four corners are adjacent to each other.
- Loops of 1, 2, 4, 8 or 16 cells eliminate 0 to 4 variables.

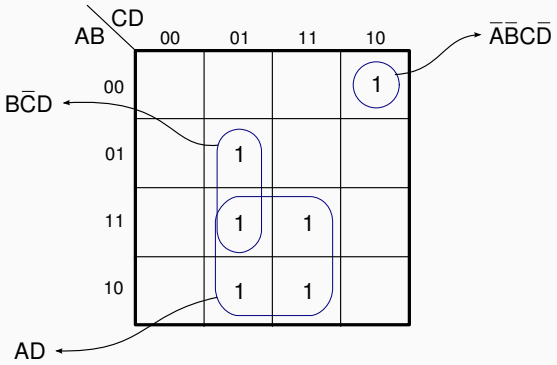
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Four-Variable Loops



- A few of the possible coverings of a four-variable map.

Example: $F = \sum m(2, 5, 9, 11, 13, 15)$



- $m_9, m_{11}, m_{13}, m_{15}$: $A = 1,\ D = 1$, term AD.
- m_5 with m_{13}: term $B\overline{C}D$.
- m_2 has no neighbour: $\overline{A}\,B\overline{C}\overline{D}$.

Result

$F = AD + B\overline{C}D + \overline{A}\,B\overline{C}\overline{D}$

Example: Wraparound at the Corners

$AB\backslash CD$	00	01	11	10
00	1	0	0	1
01	0	0	0	0
11	0	0	0	0
10	1	0	0	1

$$F = \sum m(0, 2, 8, 10)$$

- Rule 7: left and right columns are adjacent, as are top and bottom rows.
- The four corners touch and form one group of four.
- $B = 0$ and $D = 0$ in all four cells; A and C take both values and cancel.

Result

$F = \overline{B}\,\overline{D}$

Without wraparound: four isolated 1s and four four-literal terms.

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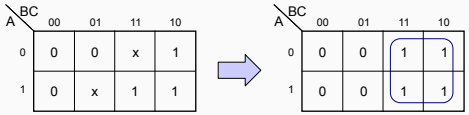
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Don’t-Care Cells

- Some inputs never occur, or their output does not matter: don’t cares, drawn as $\times$.
- A don’t care may be treated as 1 if it enlarges a group; otherwise treat it as 0 and ignore it.
- Covering a don’t care is optional.

$F = \sum m(2, 6, 7) + \sum d(3, 5)$

Set $m_3 = 1$ and $m_5 = 0$: the loop m_2, m_3, m_6, m_7 gives $F = B$.



Example: Using Don’t Cares

$AB \backslash CD$	00	01	11	10
00	0	1	1	0
01	0	×	1	0
11	0	×	1	0
10	0	×	1	0

$$F = \sum m(1, 3, 7, 11, 15) + \sum d(5, 9, 13)$$

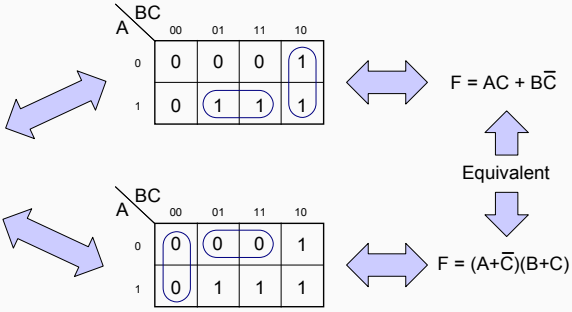
- 1s only: $F = CD + \overline{A}\,\overline{B}D$ (6 literals).
- Treat the don’t cares as 1s: cells 1, 3, 5, 7, 9, 11, 13, 15 are exactly the eight cells with $D = 1$.
- One group of eight drops three variables.

Result

$F = D$

A don’t care that does not enlarge a group is left as 0.

K-Maps for Maxterms (POS)



- Use the 0 cells. Write a sum term per group, with complemented/uncomplemented roles swapped.
- SOP: $F = AC + B\overline{C}$.
- Zeros M_1, M_3: $(A + B + \overline{C})(A + \overline{B} + \overline{C}) = A + \overline{C}$.
- Zeros M_0, M_4: $(A + B + C)(\overline{A} + B + C) = B + C$.

Result

$F = (A + \overline{C})(B + C)$

Back to the Opening Question

Why Gray-code order?

In Gray-code order, cells that are neighbours on the map differ in exactly one variable. That is what lets a group of adjacent 1s drop a variable. In plain binary order, neighbours such as 01 and 10 differ in two variables, so the unifying theorem would not apply.

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Summary

- Adjacent cells differ in one variable. Grouping 2^k adjacent 1s eliminates k variables.
- Use the largest groups and the fewest groups.
- An essential prime implicant covers a 1 no other prime implicant covers and must appear in every minimum SOP.
- Use don't-cares when they allow larger groups; covering them is optional.
- For POS, group the 0s and write a sum term for each group.

Looking ahead

Chapter 7 applies K-maps to complete combinational design problems.